

N M EMRAN HUSSAIN

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WORK STATUS: [Work permit \(EAD\) Issued](#) | Available Immediately | Relocation available.

SUMMARY

Business Data Scientist and MSBA graduate specializing in Generative AI systems, customer lifetime value (CLV) modeling, and A/B testing. Proven ability to translate complex NLP pipelines and market sizing algorithms into data-backed go-to-market strategies and scalable revenue opportunities.

TECHNICAL SKILLS

Languages & Core Frameworks: Python (Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow, Keras, PyTorch), R (tidyverse, ggplot2), SQL

AI & Generative Systems: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformers, LangChain, Vector Databases (ChromaDB), NLP (Word2Vec, FastText, Sentiment Analysis)

Data Engineering & Cloud: PySpark, Apache Spark, PostgreSQL, ETL Pipelines, Google Cloud Platform (GCP), Git/GitHub

Business Strategy & Advanced Analytics: A/B Testing, Customer Lifetime Value (CLV), Churn Optimization, Explainable AI (XAI, SHAP), ROI Analysis, Stakeholder Management

SELECTED PROFESSIONAL & PROJECT EXPERIENCE

[Community Dreams Foundation](#) | **Business Data Scientist (Volunteer)** | July 2026 – Present

Project: Elpsan Bio (Healthcare Analytics & Predictive Modeling)

- Modeled revenue potential and market adoption for 15+ FDA drug candidates across 4 disease cohorts using predictive market-sizing algorithms.
- Translated NLP models into automated stakeholder dashboards, accelerating go-to-market data retrieval by 15–20% for the research team.
- Evaluated clinical datasets to model high-ROI therapeutic targets, programmatically filtering hundreds of pharmacological matches down to 5–10 high-viability candidates.

[Predictive-RAG Retention Engine](#) | **Gemini 2.5 Flash, Reinforcement Learning, PySpark, ChromaDB, LangChain, GCP**

- Engineered an AI retention architecture utilizing LinUCB Bandits and Gemini 2.5 Flash within Google Cloud Platform.
- Deployed a Reinforcement Learning and RAG-based pipeline using PySpark, LangChain, and ChromaDB to optimize enterprise decision support and mitigate customer churn.

[NLP Sentiment Analysis Optimization](#) | **FastText, Python, PySpark, Autotune**

- Designed a distributed NLP sentiment pipeline using PySpark and FastText to process and normalize 25,000 text records.
- Optimized supervised learning models with N-grams, driving classification accuracy to 87.4% for production-ready consumer insights.

[Fairness-Aware Mortgage Pricing Model](#) | **Explainable Boosting Machines (EBM), SHAP, H2O.ai, Responsible AI**

- Built Explainable Boosting Machines (EBM) for mortgage pricing, balancing predictive performance (0.78 AUC) with federal Fair Lending compliance.
- Implemented Adversarial Impact Ratio (AIR) analysis and extraction attacks to validate model security and algorithmic fairness.

EDUCATION

The George Washington University, School of Business | Washington, DC

MS in Business Analytics (MSBA) | December 2025

Relevant Courses: Responsible Machine Learning, Practical Artificial Intelligence, Working with Large Dataset (Big Data), AI in Marketing, Customer Analytics, Text Analytics (NLP).

North South University, School of Business | Dhaka, Bangladesh

Master of Business Administration (MBA) | May 2023

Relevant Courses: Marketing Research, Brand Management, Global Marketing, Marketing Strategy, Marketing Analytics.

PROFESSIONAL EXPERIENCE

AMPERE LTD. | Founder & Data Analytics Lead | Dhaka, Bangladesh | January 2019 – December 2019

- Engineered Python and R regression models to forecast commercial demand, generating a 9% revenue lift and reducing overstock by 5%.
- Automated cost-tracking data pipelines to monitor pricing variances, successfully protecting 10% operating margins.

ANRHUS & SHUZOS CO. | Assistant Manager | Dhaka, Bangladesh | June 2009 – December 2018

- Spearheaded localized data tracking workflows to optimize infrastructure logistics and accelerate asset time-to-revenue.
- Streamlined inventory analytics and audit reporting, cutting overhead expenses by 3% and boosting operational efficiency by 4%.

CERTIFICATION

Google Professional Machine Learning Engineer (In Progress)